Math + Code Audit Report

Gross & Chivers (2025) "NIMBYism and the Housing Supply"

February 2026

This document reports the findings of a static code audit of `Code/SteadyState/` (MATLAB) and `Code/Data/` (Stata) against the paper's stated model, equations, and calibration table. Seven MATLAB files and six Stata do-files were read in full. No compile or run tests were executed.

Scope note. *`Code/Codes_ABB/` (quantitative results, impulse responses, calibration routines) was not covered in this pass and should be audited separately. The `Code/SteadyState/` folder is the likely production codebase; see Outstanding Item O1.*

Summary of findings.

Severity	MATLAB	Stata	Total
Critical	4	2	6
High	6	3	9
Medium	3	3	6
Low	2	2	4
Total	15	10	25

High-Level Math Audit (Paper First)

This section checks internal model logic and equation coherence in the paper before the code crosswalk. Source checked: `Gross and Chivers (2025) NIMBYism and the Housing Supply.lyx`.

ID	Severity	Description
HLM1	High	Median-voter regularity conditions not fully justified
HLM2	Medium	Mixed marginal and finite-difference voting objects
HLM3	Medium	Time/notation switching for prices and assets
HLM4	Medium	Calibration prose mismatches listed parameter vector
HLM5	Low	Math-adjacent typos in exposition

HLM1: Median-voter regularity conditions

[HIGH] The paper uses the equilibrium condition

$$\int g(\mathbf{s}; p_t^h)\, \Theta(\mathbf{s}; p^h)\, ds = 0.5,$$

while also noting that household preferences over house prices are not monotonic for some households. That can violate single-peakedness assumptions typically used to justify a one-dimensional median-voter equilibrium.
Inference. This is not automatically a model error, but the paper should state whether equilibrium is claimed as a numerical root condition only, or whether existence/uniqueness follows from an explicit monotonicity assumption.

HLM2: Mixed marginal and finite-difference objects

[MEDIUM] The narrative refers to marginal effects ($\partial V/\partial p$), while the formal voting rule uses:

$$V_j(\mathbf{s}; p^h + \epsilon^p) - V_j(\mathbf{s}; p^h) + \sigma \geq 0.$$

These are equivalent only under a local-limit interpretation ($\epsilon^p \to 0$) or additional smoothness assumptions.
Requested clarification. State whether ϵ^p is an infinitesimal perturbation used numerically, or a finite policy step.

HLM3: Time/notation consistency

[MEDIUM] The paper alternates between p_t and p_t^h for house price, and uses both static and recursive constraint forms (a_t and a_{t+1} contexts). This likely reflects exposition choices rather than a structural inconsistency, but it obscures timing.
Requested clarification. Add one compact timing-convention paragraph (start-of-period states, controls, end-of-period transitions), and normalize price notation.

HLM4: Calibration prose vs parameter vector

[MEDIUM] The calibration sentence describes a narrower parameter class, but the listed calibrated vector includes $\beta, \sigma, \zeta, \bar{w}, \psi, \phi^{rent}$, which spans preferences, voting, bequests, and rental-sector terms.
Requested clarification. Rewrite the sentence to classify all calibrated parameters by block.

HLM5: Minor math-adjacent typos

[LOW] There are low-risk typographic issues in formal section headers and nearby prose (e.g., “Medan Voter Theorem”). These do not alter equations but reduce presentation quality.

High-Level Math Verdict

No clear algebraic sign error was found in the core household Bellman setup, tenure-choice decomposition, or rental zero-profit equation in this high-level pass. The main outstanding math risk is the equilibrium-voting regularity argument (HLM1), followed by notation/timing clarity (HLM2–HLM4).

Critical Issues (M1–M4; S1–S2)

M1: Housing weight in utility function

[CRITICAL] The paper (Eq. 2) specifies a CES period utility function:

$$u(c_t, h_t) = \frac{\left(c_t^{1-\zeta} h_t^{\zeta}\right)^{1-\gamma}}{1-\gamma}, \qquad \zeta = 0.75,$$

so housing receives weight $\zeta = 0.75$ and consumption receives $1 - \zeta = 0.25$.

The primary solver `SolveSS.m` (line 15) and `SolveSS_iter.m` (line 14) implement:

```
omega = 0.5
value = (C^omega * (H + theta_r*R)^(1-omega))^(1-eta) / (1-eta)
```

Here `omega` is the *consumption* weight, so the housing weight is $1 - \omega = 0.50$. The correct code translation of $\zeta = 0.75$ would be `omega = 0.25` (not 0.50). Both the parameter value and the role of the symbol are reversed relative to the paper.

Impact. The consumption–housing substitution ratio is entirely determined by ζ. A value of 0.50 rather than 0.75 substantially changes optimal housing demand, ownership rates, and the price–income ratio—all calibration targets.
Decision needed. Confirm the calibrated value of ζ. If $\zeta = 0.75$ is correct, set `omega = 1 - 0.75 = 0.25`.

M2: Discount factor β

[CRITICAL] The paper's calibration table states $\beta = 0.978$.

File	Line	Value
`SolveSS.m`	17	`beta = 0.80`
`SolveSS_function.m`	16	`beta = 0.98`
`SolveSS_iter.m`	16	`beta = 0.80`
`SolveSS_Loop.m`	20	`beta = 0.80`

Only the optimisation wrapper `SolveSS_function.m` is close to the calibrated value. The remaining three files—including the main standalone solver—use $\beta = 0.80$, which implies implausibly low patience and would dramatically compress life-cycle saving and housing accumulation.
Decision needed. Identify the production file and update it to $\beta = 0.978$.

M3: Non-financial preference parameter σ absent from code

[CRITICAL] The paper defines the individual voting indicator (Eq. 15) as:

$$\Theta(s;\, p^h) = \mathbf{1}\left\{V_j(s;\, p^h + \varepsilon) - V_j(s;\, p^h) + \sigma \geq 0\right\}, \qquad \sigma = -2.4.$$

The parameter σ captures non-financial motives for voting (congestion, ideology) and is calibrated to shift approximately 15% of voters toward supply.

No variable named `sigma` or equivalent with value -2.4 appears in any `Code/SteadyState/` file. The voting block in `SolveSS.m` (lines 393–474) computes:

```
vote = sign(valuefunction_dp - valuefunction) * population_density
```

This captures only the financial component; the σ shift is absent entirely.
Impact. Without σ, the voting equilibrium is determined solely by financial incentives and will produce a different equilibrium price from the paper.
Decision needed. Confirm whether σ was intentionally dropped (e.g. absorbed into a price normalisation) or is missing. If missing, add as an additive term in the voting threshold.

M4: Rental price hardcoded; zero-profit condition absent

[CRITICAL] The paper derives the rental price from a firm zero-profit condition (Eq. 12):

$$p_t^{\text{rent}} = \varphi^{\text{rent}} + r_t^a \cdot p_t^h - \Delta p_{t+1}^h,$$

where $\varphi^{\text{rent}} = 0.05$, $r_t^a = 0.02$, and Δp_{t+1}^h is expected house price appreciation.
`SolveSS.m` (line 13) instead sets:

```
r_price = 0.09
```

This is a fixed constant, not derived from the equilibrium condition. No file in `Code/SteadyState/` computes `r_price` as a function of `a_price`, `rbPos`, or expected appreciation.
Impact. The rental price does not respond to changes in the house price or interest rate, breaking the arbitrage between the rental and ownership markets.
Decision needed. Either implement Eq. 12, or document that 0.09 is a pre-computed equilibrium value at the baseline calibration and explain why dynamic updating is unnecessary.

S1: All Stata merges drop `_merge` without validation

[CRITICAL] Every merge in the data pipeline immediately discards the match indicator without tabulating match rates. Examples from `3 birthrates.do`:

```
merge m:1 statefips year using birthrates_updated   // line 25
drop _merge                                          // line 26

merge m:m metarea using weights, keepusing(sumcity) // line 41
drop _merge                                          // line 42
```

The same pattern appears in files 5 and 6. There is no `tab _merge`, `assert _merge == 3`, or any check for unmatched observations.
Impact. Any silently dropped metropolitan areas or years will not appear in the regression sample without re-running and inserting diagnostics. This is a replication risk.
Fix. Add `tab _merge` before every `drop _merge`. For key merges, add `assert _merge == 3` or document the expected match rate.

S2: Impossible logical condition in `2 citystateweights.do`

[CRITICAL] The manual override block (around line 1234) contains at minimum one condition that is logically impossible:

```
drop if metarea == 60 & bpl == 13 & bpl == 45
```

A variable cannot simultaneously equal 13 and 45. This evaluates to false for all observations and silently does nothing. The probable intent was:

```
drop if metarea == 60 & (bpl == 13 | bpl == 45)
```

Impact. The intended observations are not dropped, potentially leaving outlier or erroneous weight cells in the shift-share instrument. This affects weighted birth rate construction and all downstream regressions.
Fix. Review the full 600-line block for similar errors and replace `&` with `|` wherever `bpl` conditions are clearly alternatives.

High Priority Issues (M5–M10; S3–S5)

M5: Retirement age

[HIGH] The paper states working life $J^{\text{ret}} = 41$ periods (ages 25–65). `SolveSS.m` (line 34) sets `agepension = 60`, implying retirement at age 60 and only 35 working periods—five years shorter than the paper. This affects the earnings profile, savings accumulation, and the age at which households transition from renting to owning.

M6: Risk-free savings rate

[HIGH] The calibration table states $r^a = 0.02$ (2%). `SolveSS.m` (line 10) sets `rbPos = 0.03` (3%). This affects all saving decisions, the rental price via Eq. 12, and the mortgage spread. With the paper's spread of 3 percentage points, a savings rate of 3% implies a mortgage rate of 6%, not 5%.

M7: Bequest parameters

[HIGH] The paper (Eq. 4 and calibration table) specifies:

$$v(w) = \psi \, \frac{(w + \bar{w})^{1-\gamma}}{1-\gamma}, \qquad \psi = 10, \quad \bar{w} = 150.$$

`SolveSS.m` (lines 23–24) sets `bequestweight1 = 0.72` and `bequestweight2 = 0.72`, and implements:

```
bequestweight1 * (1 + Bequest/bequestweight2)^(1-eta)
```

Neither value is remotely close to $\psi = 10$ or $\bar{w} = 150$. The normalisation $(1 + B/\bar{w})$ could be algebraically equivalent to $(w + \bar{w})^{1-\gamma}$ up to a scaling constant, but `bequestweight2` $= 0.72 \neq 150$.
Decision needed. Provide the algebraic equivalence mapping both parameter sets, or correct the values.

M8: Transaction cost inconsistent across files

[HIGH] The paper states $F^{\text{sell}} = 0.07$ (7%).

File	Line	`ka`
`SolveSS.m`	22	0.07 ✓
`SolveSS_function.m`	18	0.06
`SolveSS_Loop.m`	22	0.09

Two of three function files use a different value from the paper. F^{sell} enters the budget constraint for adjusting owners (Eqs. 7 and 10) and governs the margin between adjusting and non-adjusting tenure.

M9: Rental discount θ^{rent} inconsistent across files

[HIGH] The calibration table (following Mian et al. 2017) states $\theta^{\text{rent}} = 0.90$.

File	Line	theta_r
SolveSS.m	16	0.90 ✓
SolveSS_function.m	15	0.85
SolveSS_Loop.m	19	0.85

θ^{rent} governs the utility discount for renting vs. owning and directly shifts the rent–own margin.

M10: Utility form inconsistent across solver files

[HIGH] The paper specifies CRRA utility with $\gamma = 2$ (Eq. 2). The files differ:

File	Utility form
SolveSS.m	CRRA, $\eta = 2$
SolveSS_iter.m	CRRA, $\eta = 2$
SolveSS_function.m	Log: $\log c + s_h \log(h + \theta_r R)$
SolveSS_Loop.m	Log

Log utility is the limiting case $\gamma \to 1$ and constitutes a different model. If `SolveSS_Loop.m` is the grid-search production solver, the equilibrium prices it generates will not correspond to the CRRA model described in the paper.

S3: 600+ lines of manual `replace share = 0` with no documentation

[HIGH] `2 citystateweights.do` lines 615–1234 contain over 600 statements of the form:

```
replace share = 0 if metarea == X & bpl == Y
```

No comment explains the selection criterion. The logic is hand-coded, not rule-based (e.g. `replace share = 0 if count < 5`).
Impact. The shift-share instrument is constructed from these weights. Undocumented exclusions affect instrument validity and cannot be assessed by a reader or replicator.
Recommendation. Replace the manual block with a programmatic rule and document the threshold. If individual overrides are genuinely case-specific, add a comment to each.

S4: Hardcoded machine-specific file paths

[HIGH] Multiple files contain paths tied to specific machines:

File	Line	Path
1 tenurechoices.do	1	/Users/igro0002/Dropbox/…
4 importexcel .do	2, 13	/Users/igro0002/Dropbox/…
merge data.do	6–7	C:\Users\Dave_\…

The mixture of macOS and Windows paths indicates the pipeline was developed across machines without standardisation. No collaborator or replicator can run the code without modifying these lines.
Fix. Replace all hardcoded paths with a single global macro at the top of each file, e.g.:

```
global datadir "C:\Users\Dave_\Dropbox\Zac and David\Data"
use "$datadir/usa_00009.dta"
```

S5: IV regressions without instrument diagnostics

[HIGH] `5 combinepermits.do` (lines 103–118) runs 50 IV regression specifications in nested loops. No first-stage F-statistics are reported and no Kleibergen–Paap or Cragg–Donald statistics are computed. The outlier exclusion `abs(totalunits[h]growth) < 2` is not documented.
Recommendation. Add `estat firststage` to each IV loop and export first-stage F-statistics alongside point estimates in `Regressions.csv`.

Medium Priority Issues (M11–M13; S6–S8)

M11: Housing grid inconsistent across files

[MEDIUM] The paper does not report the grid size used for published results. The files vary substantially:

File	J	`housingmax`	Liquid grid I
`SolveSS.m`	40	14	50
`SolveSS_function.m`	10	15	50
`SolveSS_iter.m`	20	25	50
`SolveSS_Loop.m`	20	30	60

A grid of $J = 10$ is very coarse and likely insufficiently accurate for value function approximation. The varying `housingmax` changes which housing quantities are accessible to households. Document which grid was used for the published results.

M12: `agemax = 90` in `SolveSS_function.m`

[MEDIUM] The paper states agents live from age 25 to age 80 ($J = 56$ periods). `SolveSS_function.m` (line 26) sets `agemax = 90`, extending the model life by 10 years. With `dage = 5` in that file (coarser age grid), the terminal condition and bequest calculation apply at a different age than stated in the paper.

M13: Notation reversal between paper and code

[MEDIUM] The paper uses ζ as the *housing* weight ($\zeta = 0.75$, so housing receives 75%). The code uses `omega` as the *consumption* weight. The correct translation is `omega = 1 - zeta = 0.25`. The code instead uses `omega = 0.50` (see M1). The reversal in naming conventions creates a risk of silent errors whenever the utility function is re-parameterised.

S6: Lag construction duplicated between files 3 and 5

[MEDIUM] `3 birthrates.do` constructs lags `l1bw–l50bw` (lines 55–119). `5 combinepermits.do` reconstructs `l1bw–l65bw` (lines 13–77). If the panel changes between files due to the crosswalk merge, lags may be computed over a different `tsset` structure. Any change to the lag range in file 3 must be replicated manually in file 5.

S7: `statefips / 100` conversion

[MEDIUM] `3 birthrates.do` (line 21) converts IPUMS birth-state codes to FIPS codes via `replace statefips = statefips / 100`. This arithmetic conversion may fail for codes that are not round multiples of 100 (e.g. District of Columbia at IPUMS code 98 → 0.98, not a valid FIPS code).

Recommendation. Use the IPUMS-provided state crosswalk rather than division. Verify all resulting values are valid FIPS codes before merging.

S8: Predicted variable created but unused

[MEDIUM] `5 combinepermits.do` (line 99) creates `own_predicted` after a fixed-effects regression but this variable is never used in any subsequent command and is not saved to output. Remove or document its purpose.

Low Priority Issues (M14–M15; S9–S10)

M14: `IncomeProcess.m` is an incomplete fragment

[LOW] `IncomeProcess.m` ends at line 104, mid-construction of lower-diagonal arrays for an income transition matrix. There is no return statement, no assembled sparse matrix, and no output. The file cannot be called as a standalone function. It appears to be a copy-paste fragment that was never completed.

M15: Unused variable `sigma` in `SolveSS.m`

[LOW] `SolveSS.m` (line 18) declares `sigma = 1.5` but this variable is never referenced again. The name coincides with the paper's non-financial preference parameter $\sigma = -2.4$ (see M3), but the value differs and it is unused. Either remove or rename to avoid confusion.

S9: Commented-out Wharton merge in `merge data.do`

[LOW] Lines 28–32 of `merge data.do` contain a commented-out merge with a Wharton land-use regulation dataset (WRLURI), listed in the paper as a potential instrument for housing supply. No explanation is given for why it was removed. Document the decision.

S10: Age bin boundary overlap

[LOW] `1 tenurechoices.do` creates `age20_30` for ages 20–30 and `age30_40` for ages 30–40. Age 30 falls in both bins. This may be intentional but can cause double-counting in regression or summary tables. Confirm and document.

Confirmed Matches

The following items agree between the paper's calibration table and the primary solver `SolveSS.m`:

- [CONFIRMED MATCH] **LTV ratio** m: Paper $m = 0.90 \rightarrow$ `CC = 0.90` (line 26).
- [CONFIRMED MATCH] **Risk aversion** γ: Paper $\gamma = 2 \rightarrow$ `eta = 2` (`SolveSS.m`; `SolveSS_iter.m`).
- [CONFIRMED MATCH] **Transaction cost (primary file)**: Paper $F^{\text{sell}} = 0.07 \rightarrow$ `ka = 0.07` (`SolveSS.m` line 22).
- [CONFIRMED MATCH] **Rental discount (primary file)**: Paper $\theta^{\text{rent}} = 0.90 \rightarrow$ `theta_r = 0.90` (`SolveSS.m` line 16).
- [CONFIRMED MATCH] **Life span**: Paper $J = 56$ (ages 25–80) $\rightarrow$ `agemin = 25`, `agemax = 80`, `dage = 1` (`SolveSS.m`).
- [CONFIRMED MATCH] **Borrowing constraint form**: Paper $a_t \geq -m \cdot h_t \cdot p_t^h \rightarrow$ `-b > a*P_h*CC` (`SolveSS.m` lines 144–147).
- [CONFIRMED MATCH] **Backward induction structure**: Paper solution algorithm (backward from terminal age, forward distribution simulation) $\rightarrow$ `SolveSS.m` backward loop from `agemax` to `agemin` followed by forward pass.
- [CONFIRMED MATCH] **Voting direction**: Paper votes to restrict supply when value rises with price $\rightarrow$ `sign(valuefunction_dp - valuefunction)` (`SolveSS.m` lines 393–474). Note: financial component only; σ term absent (see M3).

Referee Recommendation (Code and Replication)

Recommendation: Major code revision before treating the current repository as a replication-ready implementation of the published model.

This recommendation is based on observed implementation mismatches (M1–M4), parameter inconsistencies across solver files (M5–M10), and missing data-pipeline diagnostics (S1–S5). As an inference, if the published quantitative results were generated from a different private branch, the immediate requirement is to document the production-file mapping and provide a clean reproducible pipeline in this repository.

Minimum acceptance checklist for replication release:

- Identify and document the exact production solver file(s) and calibration used for published tables/figures.
- Reconcile utility, discounting, voting, and rental-price equations with the paper, or explicitly document approved deviations.
- Add merge diagnostics and path standardisation to Stata scripts and re-run to regenerate key regression outputs.
- Publish a short `README_code.md` mapping each paper equation block to code locations.
- Run and archive one end-to-end reproducibility test from raw inputs to final outputs.

Outstanding Items

1. **O1 (production file)**: `Code/SteadyState/` is the likely production codebase, but `SolveSS.m` contains parameter values inconsistent with the calibration table (M1–M9). Verify whether `SolveSS.m` is the production file or a development variant; document in a `README_code.md`. *Owner: Authors.*

2. **O2 (housing weight)**: Is $\zeta = 0.75$ (housing weight) or $\omega = 0.50$ (consumption weight) the calibrated value? Reconcile code against calibration table. *Owner: Authors.*

3. **O3 (σ absent from code)**: Was $\sigma = -2.4$ intentionally dropped? If not, add as an additive term in the voting indicator before the median voter calculation. *Owner: Authors.*

4. **O4 (rental price)**: Is `r_price = 0.09` a pre-computed equilibrium value or an error? If pre-computed, document the derivation. If an error, implement Eq. 12. *Owner: Authors.*

5. **O5 (retirement age)**: Retirement age 60 (code) or 65 (paper)? Update whichever is wrong. *Owner: Authors.*

6. **O6 (bequest parameters)**: Provide algebraic equivalence between ($\psi = 10$, $\bar{w} = 150$) and (`bequestweight1` $= 0.72$, `bequestweight2` $= 0.72$), or correct the code. *Owner: Authors.*

7. **O7 (`Codes_ABB/` audit)**: The quantitative results, impulse responses, and calibration routines in `Code/Codes_ABB/` were not covered in this pass. Schedule a separate audit before resubmission. *Owner: Auditor.*

8. **O8 (impossible bpl condition)**: Review and fix the manual override block in `2 citystateweights.do`; re-run the weights and all downstream files. *Owner: Code author.*

9. **O9 (merge diagnostics)**: Add `tab _merge` before every `drop _merge` across the Stata pipeline. *Owner: Code author.*

10. **O10 (share override documentation)**: Replace or document the 600-line manual `replace share = 0` block with a programmatic rule or per-line comments. *Owner: Code author.*